

Comprehensive Post-Remission Discharge Letter

To: Dr. Evelyn Brooks (Primary Care)

From: Hematology/Oncology Service

Date: 20.04.2024

Patient: Donna Adams (*30.08.1963), MRN AML042

Primary diagnosis: Acute Myeloid Leukemia (AML), **ELN 2022**

Favorable, NPM1-mutated / FLT3-ITD negative

Initial therapy: Induction "7+3" (cytarabine + daunorubicin) **started 09.10.2021**, followed by consolidation

Reason for letter

Formal discharge from active chemotherapy with transition to surveillance care after durable complete remission and MRD negativity.

Presentation & Diagnostics

Ms. Adams presented in October 2021 with several weeks of fatigue, easy bruising, and gingival bleeding. Initial labs: **WBC $21.7 \times 10^9/L$** (peripheral blasts 32%), **Hgb 7.8 g/dL**, **platelets $46 \times 10^9/L$** , **LDH 566 U/L**. Coagulation normal; D-dimer mildly elevated. Peripheral smear showed circulating myeloblasts with Auer rods.

Bone marrow biopsy 02.10.2021: ~65% myeloblasts; flow cytometry **CD34^{low}**, **CD117⁺**, **HLA-DR⁺**, **CD13⁺**, **CD33⁺**, aberrant **CD56**.

Cytogenetics: 46,XX normal karyotype.

Molecular: **NPM1** mutation detected; **FLT3-ITD/TKD**

negative; IDH1/2, CEBPA wild type; no high-risk lesions on extended myeloid panel.

Baseline echocardiogram: LVEF 62%. Viral serologies negative.

Treatment Summary

- **Induction (7+3)** commenced **09.10.2021**. Complications included **febrile neutropenia** (cultures negative) treated with piperacillin-tazobactam, **grade 2 mucositis**, and transient liver enzyme elevation (ALT peak 84 U/L).

- **Day-28 marrow (08.11.2021): Complete remission (CR)** with <2% blasts. MRD by NPM1 qPCR: 2.1×10^{-4} , representing >3-log reduction from diagnostic baseline.
- **Consolidation:** High-dose cytarabine (HiDAC) **3 g/m² q12h D1,3,5 × 3 cycles** (Dec 2021 – Feb 2022). Neuro checks performed each cycle; no cerebellar toxicity. One re-admission in Jan 2022 for rhinovirus-associated neutropenic fever; no bacteremia.
- **MRD trajectory:**
 - Post-consolidation (03.2022): **NPM1 qPCR 3.4×10^{-5}**
 - 09.2022: **Undetectable** (assay sensitivity 10^{-5})
 - Serial peripheral blood qPCRs **undetectable** thereafter (latest **03.2024**).
- **Central venous access:** Right IJ tunneled catheter placed at induction; removed **04.2022**.

Given favorable risk and sustained MRD negativity, we **did not recommend allogeneic transplant**. We discussed maintenance therapy; after risk–benefit review, she elected **no maintenance** and proceeded with surveillance.

Current Status (clinic 15.04.2024)

Symptoms/Function: Back to full-time work (accounting). Walks 5 km on weekends. No fevers, night sweats, infections, or bleeding. ECOG performance status **0**.

Exam: Afebrile; BP 118/74; HR 70; lungs clear; no lymphadenopathy or organomegaly; no petechiae. Oral mucosa intact.

Laboratory data (15.04.2024):

- **CBC:** WBC $5.1 \times 10^9/L$ (ANC 2.6), Hgb **12.9 g/dL**, Plt **$236 \times 10^9/L$** ; MCV 90 fL.
- **CMP:** Na 139, K 4.2, Cr 0.88 mg/dL (eGFR >90), AST 21, ALT 19, Tbili 0.6.
- **Hemolysis/Inflammation:** LDH 176 U/L; CRP <1 mg/L.
- **MRD:** NPM1 qPCR **undetectable** (peripheral blood).
- **Echocardiogram (03.2024):** LVEF 61%; no change from baseline.

Imaging: None indicated. Prior chest radiographs during neutropenic episodes showed no focal consolidation.

Comorbidities & Long-Term Considerations

- **Type 2 diabetes mellitus**, diagnosed 2018; **A1c 6.8%** (03.2024) on **metformin ER 1,000 mg nightly**; no microalbuminuria.
- **Primary hypothyroidism (Hashimoto)** on **levothyroxine 75 µg qAM**; TSH 2.1 mIU/L.
- **Hyperlipidemia** on **rosuvastatin 10 mg qHS**; LDL 84 mg/dL.
- **Osteopenia** (DXA 2023 T-score -1.6 at femoral neck); on **cholecalciferol 1,000 IU daily, calcium 1,000 mg/day**, weight-bearing exercise.
- Remote **migraines without aura**; uses **sumatriptan 50 mg PRN** (warned to avoid during any future severe hypertension).
- Immunizations updated: influenza annually; COVID boosters; pneumococcal series complete.

Allergies: NKDA.

Medications on Discharge to Surveillance

- Metformin ER 1,000 mg qHS
 - Levothyroxine 75 µg qAM
 - Rosuvastatin 10 mg qHS
 - Vitamin D₃ 1,000 IU daily; calcium carbonate PRN to reach 1,000 mg/day
 - Sumatriptan 50 mg PRN migraine (≤2 doses/24 h)
 - Acetaminophen PRN pain; **avoid NSAIDs if thrombocytopenic in the future**
-

Late Effects Screening / Survivorship Plan

- **Cardiac:** Anthracycline exposure tracked; next echo **2026** or sooner if symptoms.
 - **Neuro-ocular:** No cytarabine cerebellar effects; routine eye exams annually.
 - **Fertility/gynecology:** Post-menopausal; routine screening per guidelines.
 - **Infection risk:** None beyond general population at present; advise early evaluation of fever.
 - **Iron overload:** Ferritin 240 ng/mL (03.2024); no phlebotomy indicated.
-

Follow-Up Plan

1. **Surveillance:** CBC every **3 months** through 2024, then consider spacing to **q4–6 months** if stable. **NPM1 qPCR** in peripheral blood **q6 months** for two additional years.
2. **When to call:** Temperature $\geq 38.0^{\circ}\text{C}$, new bruising/petechiae, persistent fatigue, night sweats, or unintentional weight loss.
3. **Dental/procedures:** Routine care allowed; if future thrombocytopenia recurs, coordinate hemostasis plan.
4. **Lifestyle:** Continue exercise and diabetes optimization; maintain vaccinations (non-live).

Summary: Ms. Adams remains in **ongoing complete remission** with **sustained MRD negativity** and normal blood counts after induction and consolidation. No further chemotherapy is planned, and she transitions to surveillance with our clinic.

Please contact our team with any concerns.

Warm regards,

[Name], MD — Consultant Hematologist